

PROJECT MANAGEMENT

AGENDA

- Project Plan Structure
- Activity Organization
- Project Scheduling
- Scheduling Problems
- Risk
- Risk Management

Management activities

- Proposal writing
- Project planning and scheduling
- Project costing
- Project monitoring and reviews
- Personnel selection and evaluation
- Report writing and presentations

PROJECT PLAN STRUCTURE

- Introduction
- Project Organisation
- Risk Analysis
- Hardware and Software Resource Requirements
- Work breakdown
- Project Schedule
- Monitoring and Reporting Mechanisms

Project planning process

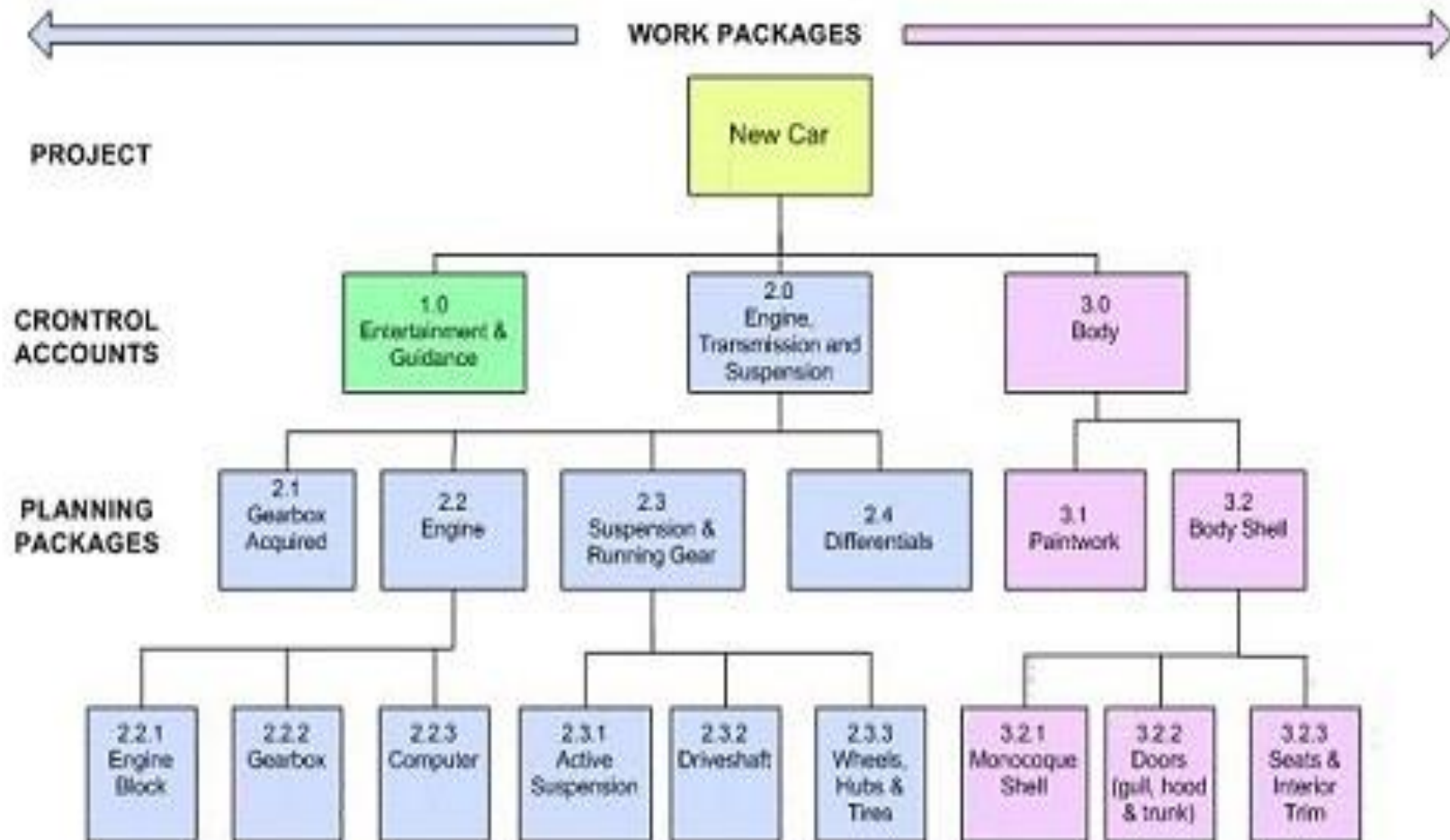
```
Establish the project constraints
Make initial assessments of the project parameters
Define project milestones and deliverables
while project has not been completed or cancelled loop
    Draw up project schedule
    Initiate activities according to schedule
    Wait (for a while )
    Review project progress
    Revise estimates of project parameters
    Update the project schedule
    Re-negotiate project constraints and deliverables
    if ( problems arise then
        Initiate technical review and possible revision
    end if
end loop
```

Work Breakdown Structure

- A **Work Breakdown Structure** (WBS) is a deliverable-oriented hierarchical decomposition of the work to be executed by the project team to accomplish the project objectives and create the required deliverables.
- All the **work** contained within the WBS is to be identified, estimated, scheduled, and budgeted.

Example

Work Breakdown Structure



Cont...

A good WBS should exhibit the following characteristics:

- Definable—can be described and easily understood by project participants.
- Manageable—a meaningful unit of work where specific responsibility and authority can be assigned to a responsible individual.
- Estimate able—duration can be estimated in time required to complete, and cost can be estimated in resources required to complete.

Cont...

- Independent—minimum interface with or dependence on other ongoing elements (i.e., assignable to a single control account, and clearly distinguishable from other work packages).
- Integratable—integrates with other project work elements and with higher level cost estimates and schedules to include the entire project.
- Measurable—can be used to measure progress; has start and completion dates and measurable interim milestones.
- Adaptable—sufficiently flexible so the addition/elimination of work scope can be readily accommodated in the WBS framework.

Tips for WBS

- Review the Work Breakdown Structure. Make sure all deliverables have been fully covered by the works defined in the Work Breakdown Structure.
- Ensure that testing and training have been taken into account.
- Ensure that non-IT work packages are also included such as, documentation and review activities are included in the structure.

Cont...

- Ensure that other supporting activities such as, product/service launch and implementation activities are planned.
- Ensure that delivery approval cycles are taken into account.
- Include project management deliverables on the project as well (e.g. production of Project Plan). Include any deliverables that must be met or delivered by the customer or any external parties. Check the Work Breakdown Structure against the project approach specified in Project Charter for any activities that needs to be included in the Work Breakdown Structure.

Phases of WBS

Phase-based structures

- Define and structure project activities based on the project phases.

Deliverable-based structures

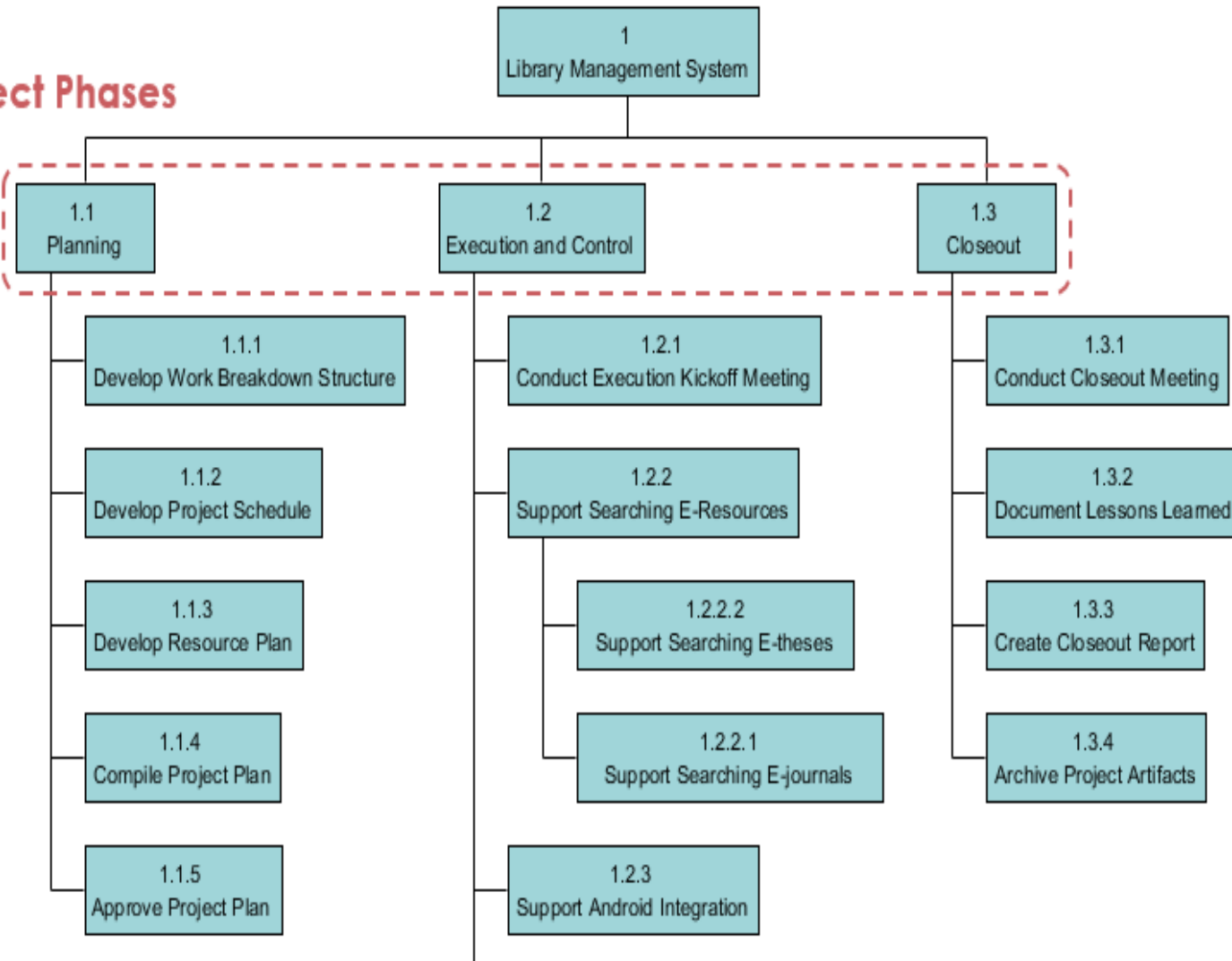
- Define and structure project activities based on the deliverables agreed to deliver.

Responsibility-based structure

- Define and structure project activities based on the organization units that will work on the project.

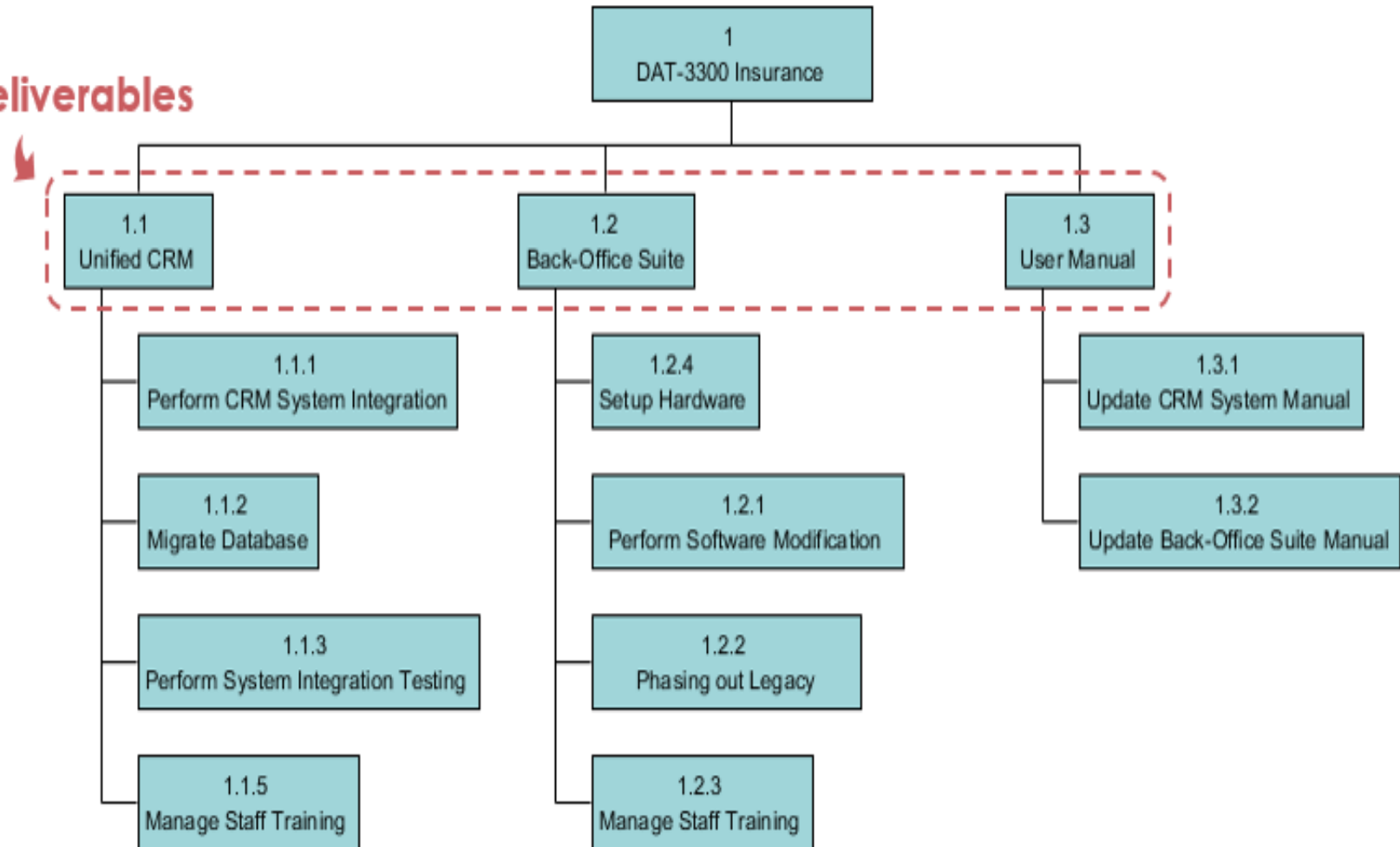
Phase-based structures

Project Phases



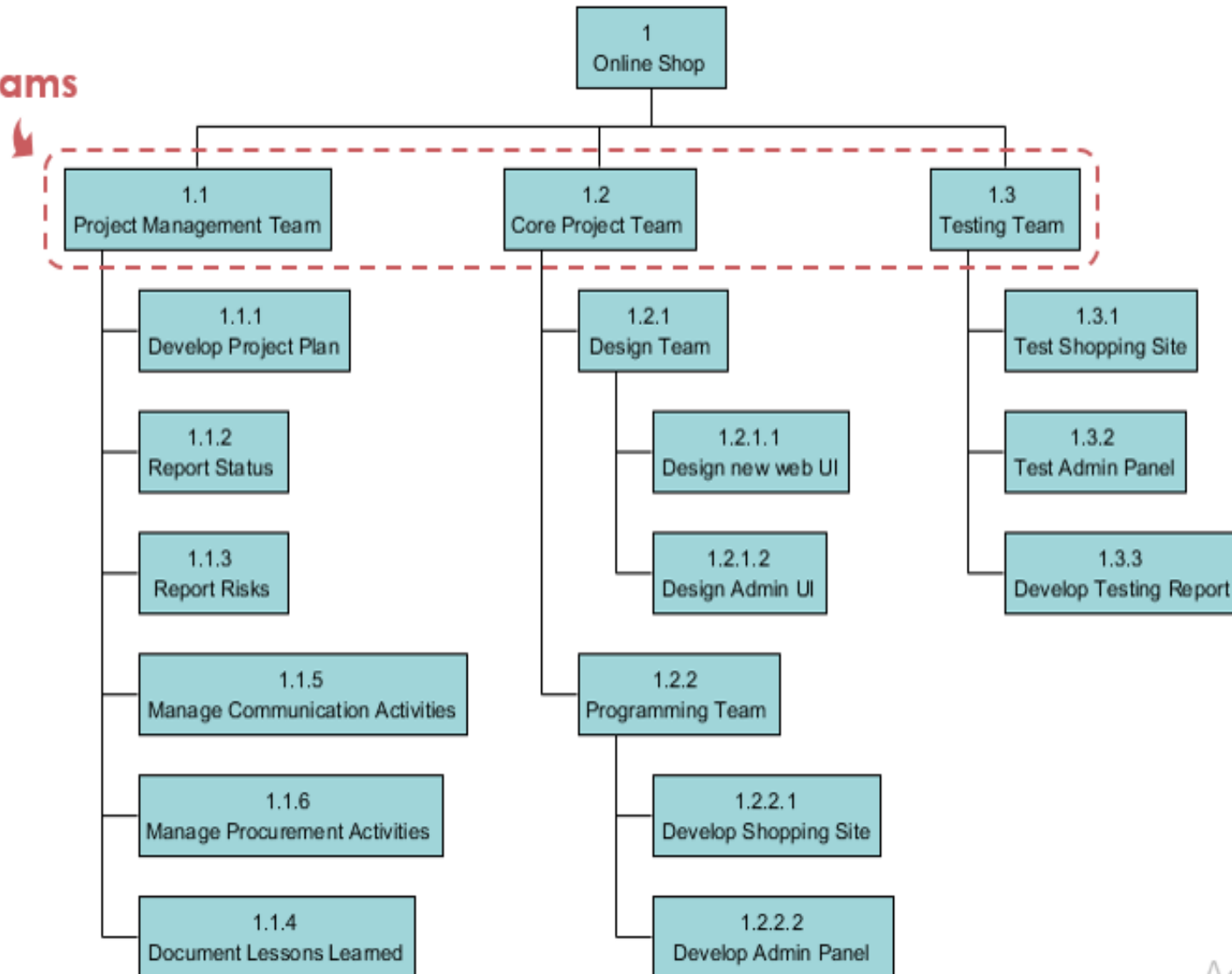
Deliverable-based structures

Deliverables

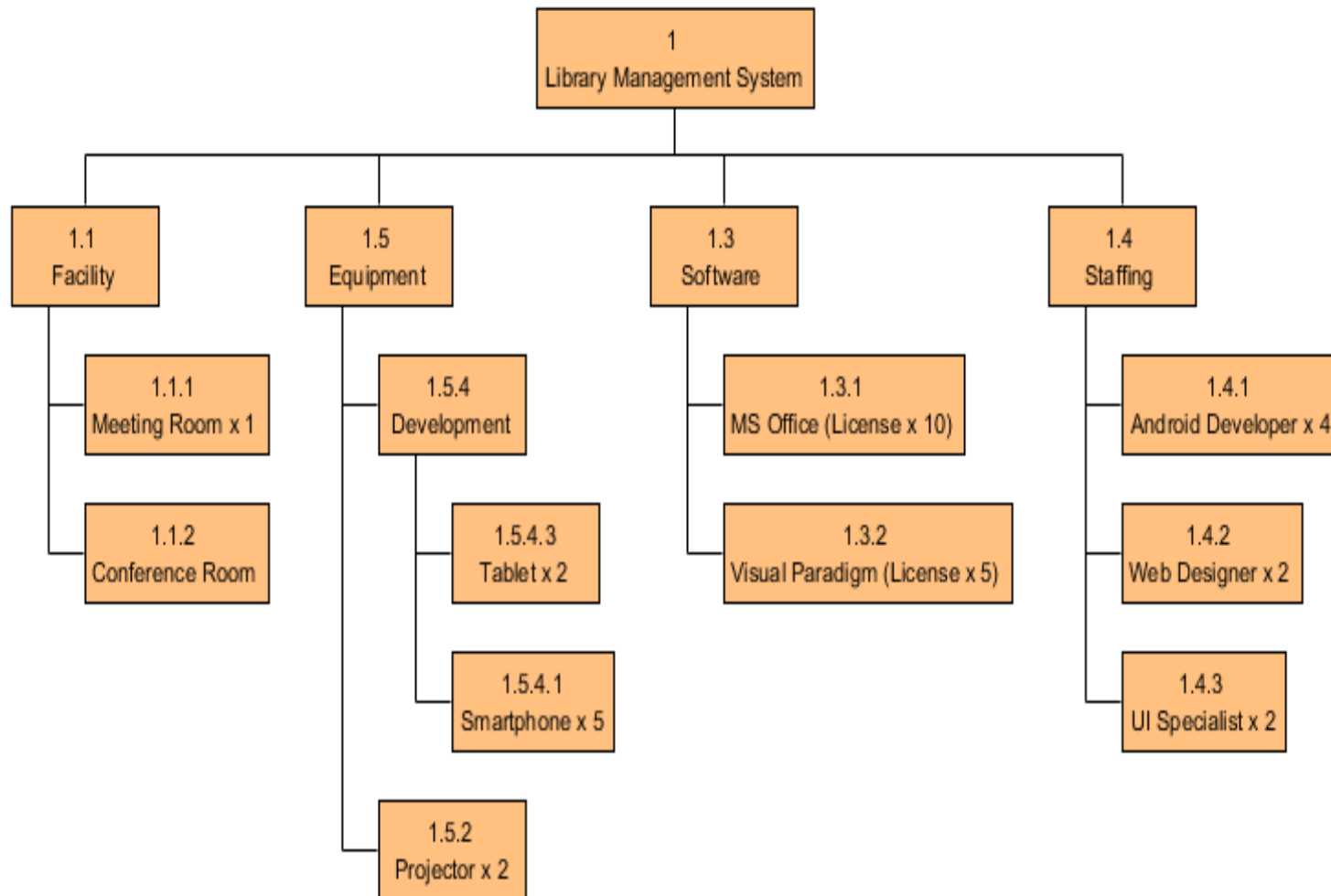


Responsibility-based structure

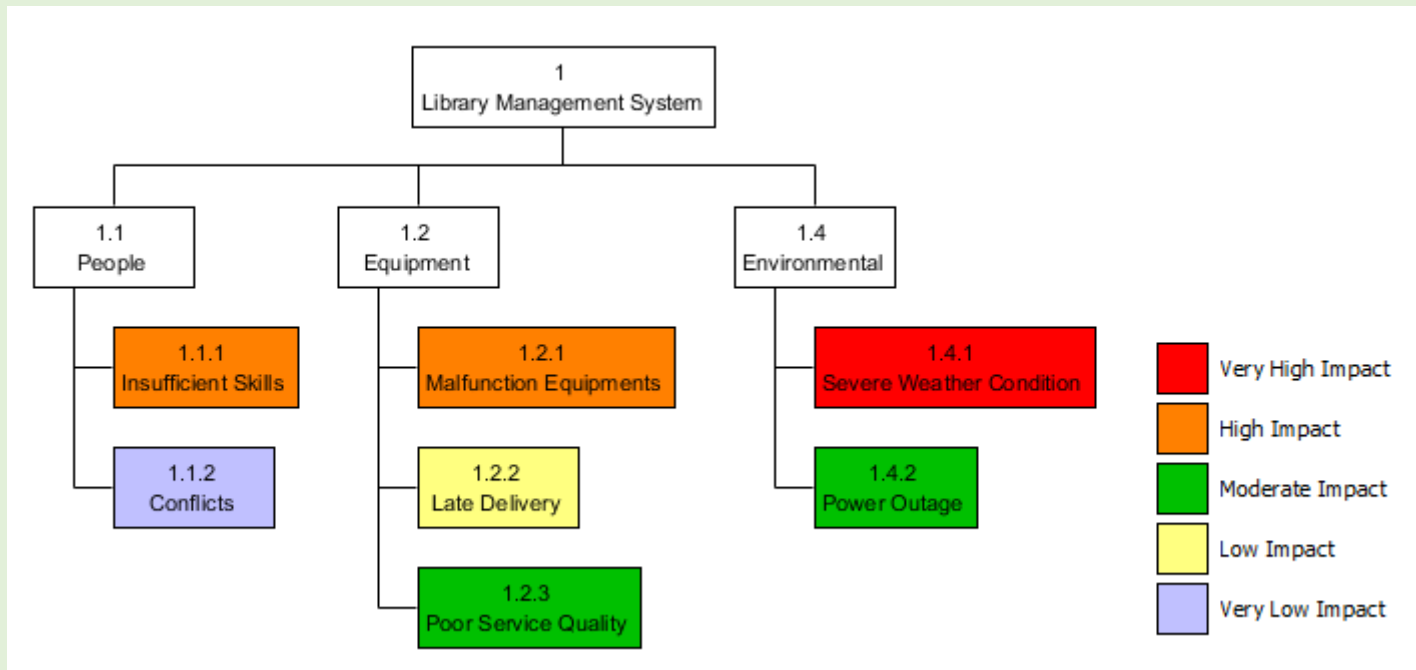
Teams



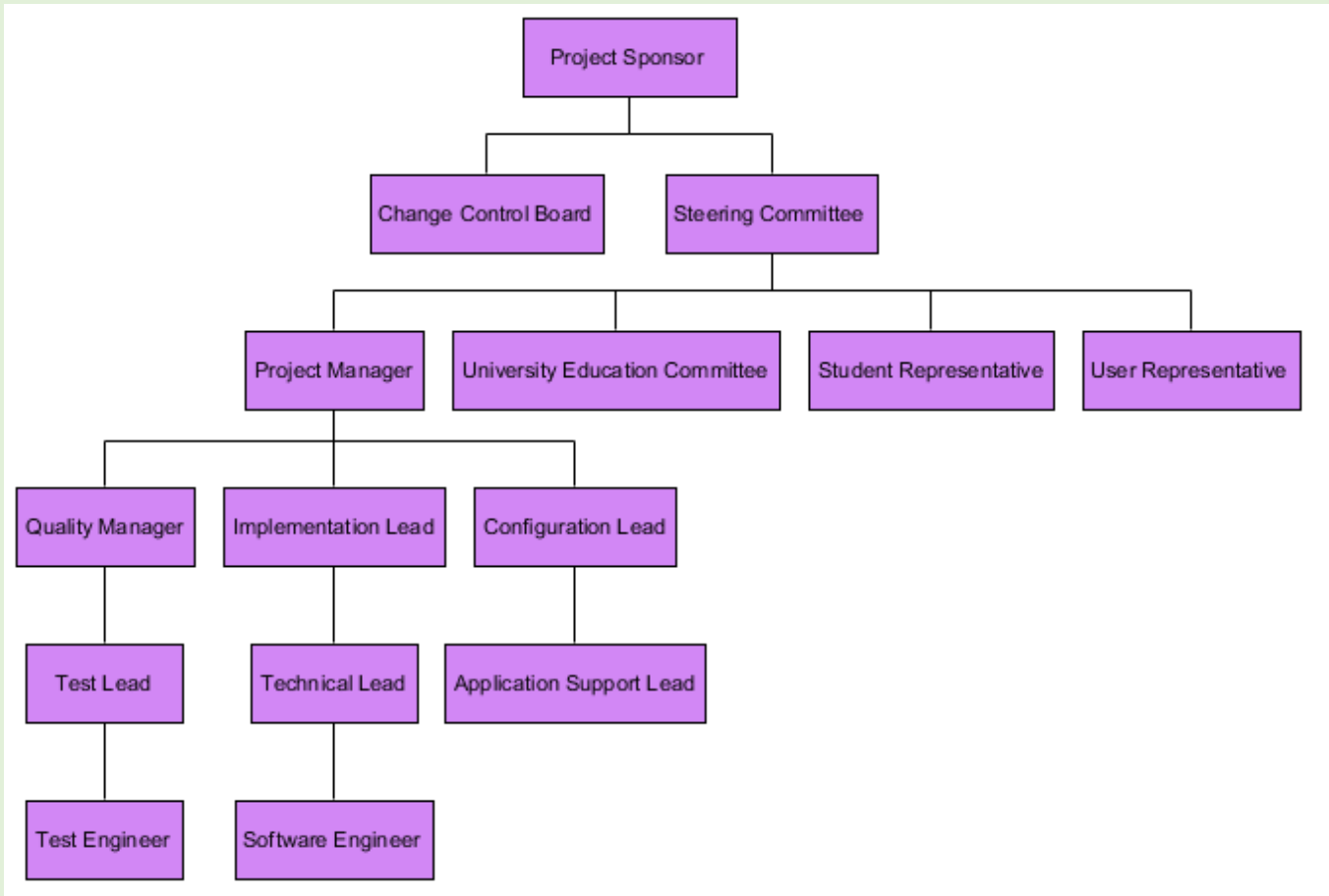
Resource Breakdown



Risk Breakdown



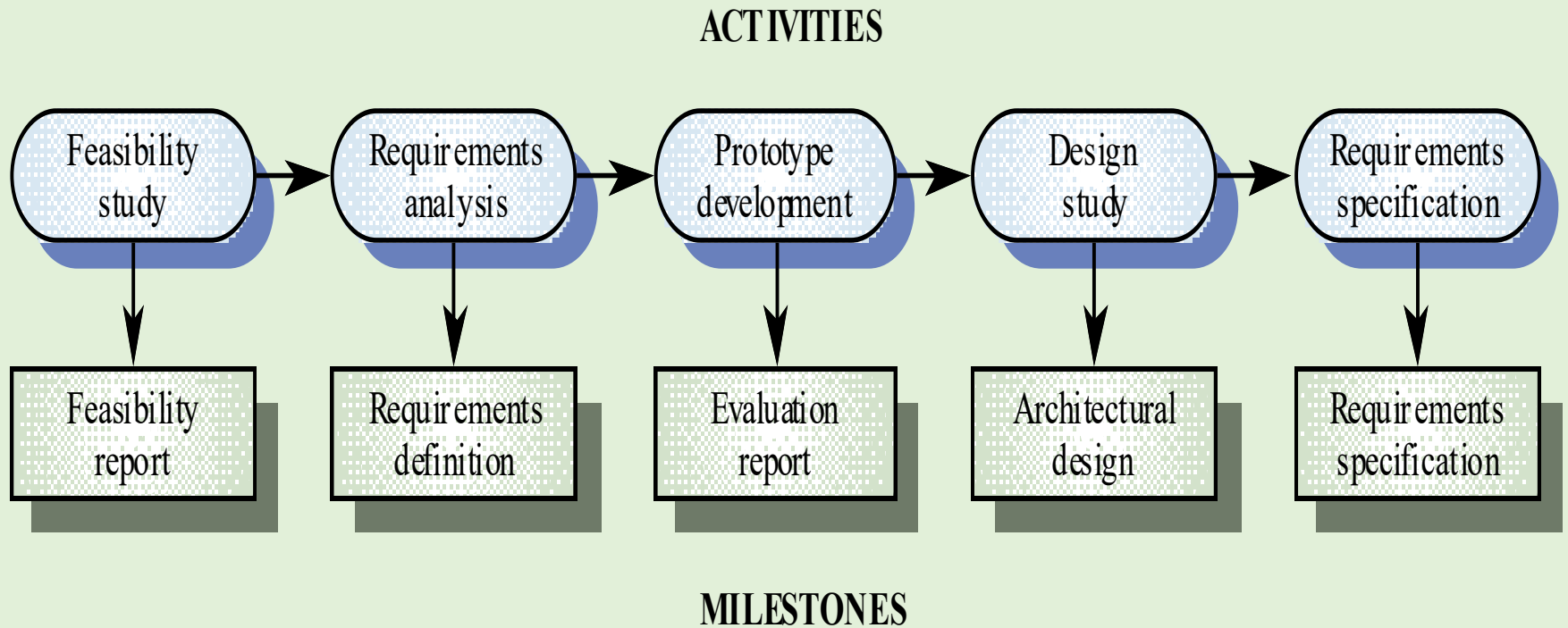
Organizational Breakdown



Activity Organization

- Activities in a project should be organised to produce tangible outputs for management to judge progress
- *Milestones* are the end-point of a process activity
- *Deliverables* are project results delivered to customers
- The waterfall process allows for the straightforward definition of progress milestones

Milestones



Project Scheduling

- Split project into **tasks** and estimate time and resources required to complete each task
- Organize tasks concurrently to make **optimal use of workforce**
- Minimize **task dependencies** to avoid delays caused by one task waiting for another to complete
- Dependent on **project managers** intuition and experience

The Project Scheduling Process

- Identify activities
- Identify activity dependencies
- Estimate resources for activities
- Allocate people to activities
- Create project charts

Scheduling Problems

- Estimating the difficulty of problems and hence the **cost** of developing a solution is hard
- **Productivity** is not proportional to the number of people working on a task
- Adding people to a **late project** makes it later because of communication overheads
- The **unexpected** always happens. Always allow contingency in planning

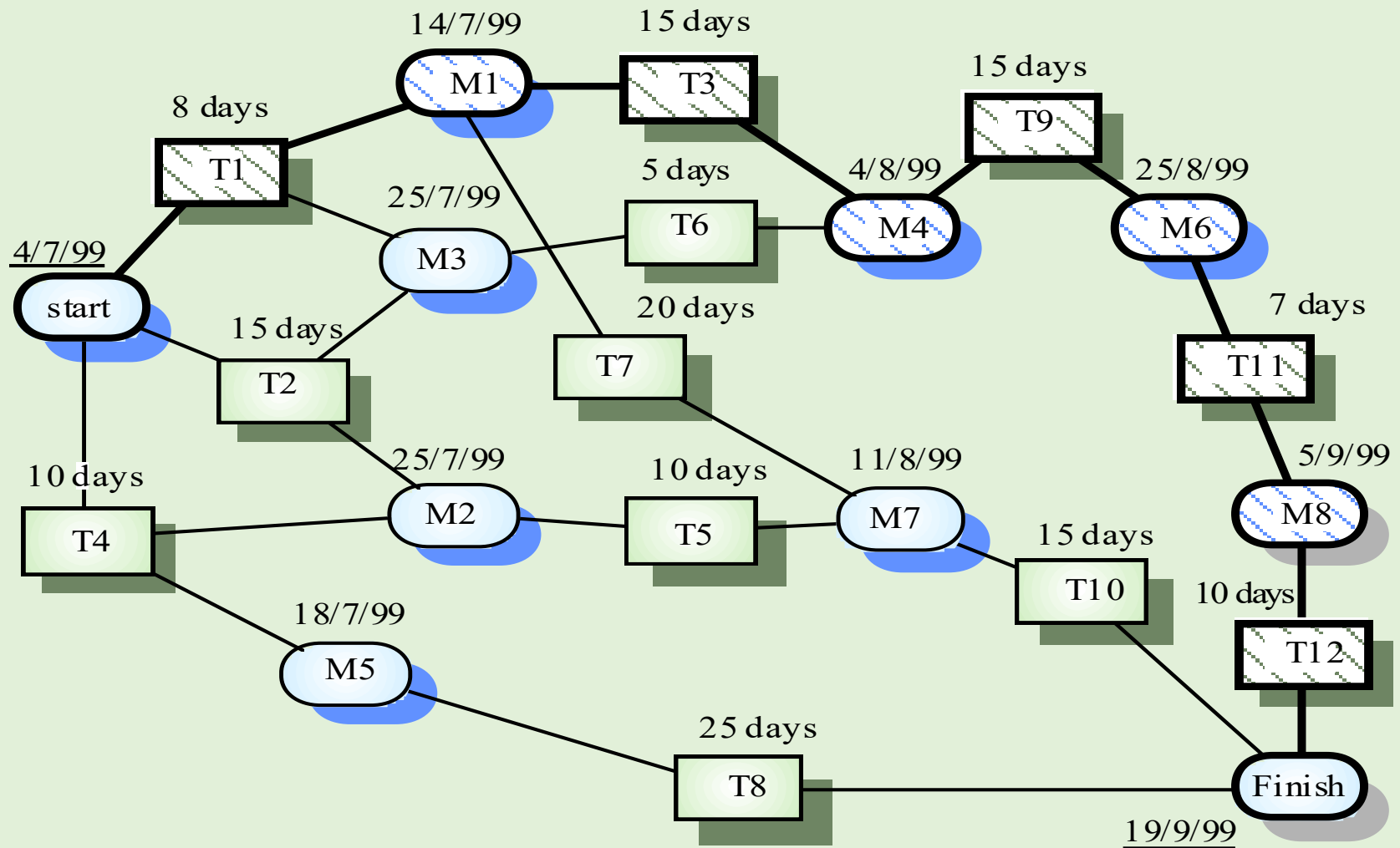
Bar Charts and Activity Charts

- Graphical notations used to illustrate the project schedule
- Show **project breakdown into tasks**. Tasks should not be too small. They should take about a week or two
- Activity charts show task dependencies and the critical path
 - the sequence of stages determining the **minimum time** needed for an operation, especially when analyzed on a computer for a large organization.
- Bar charts show schedule against calendar time

Problem

TASK	DURATION	DEPENDENCIES
T1	8 days	
T2	15	
T3	15	T1(M1)
T4	10	
T5	10	T2, T4(M2)
T6	5	T1,T2(M3)
T7	20	T1(M1)
T8	25	T4(M5)
T9	15	T3,T6(M6)
T10	15	T5,T7(M7)
T11	7	T9(M4)
T12	10	T11(M8)

Activity network



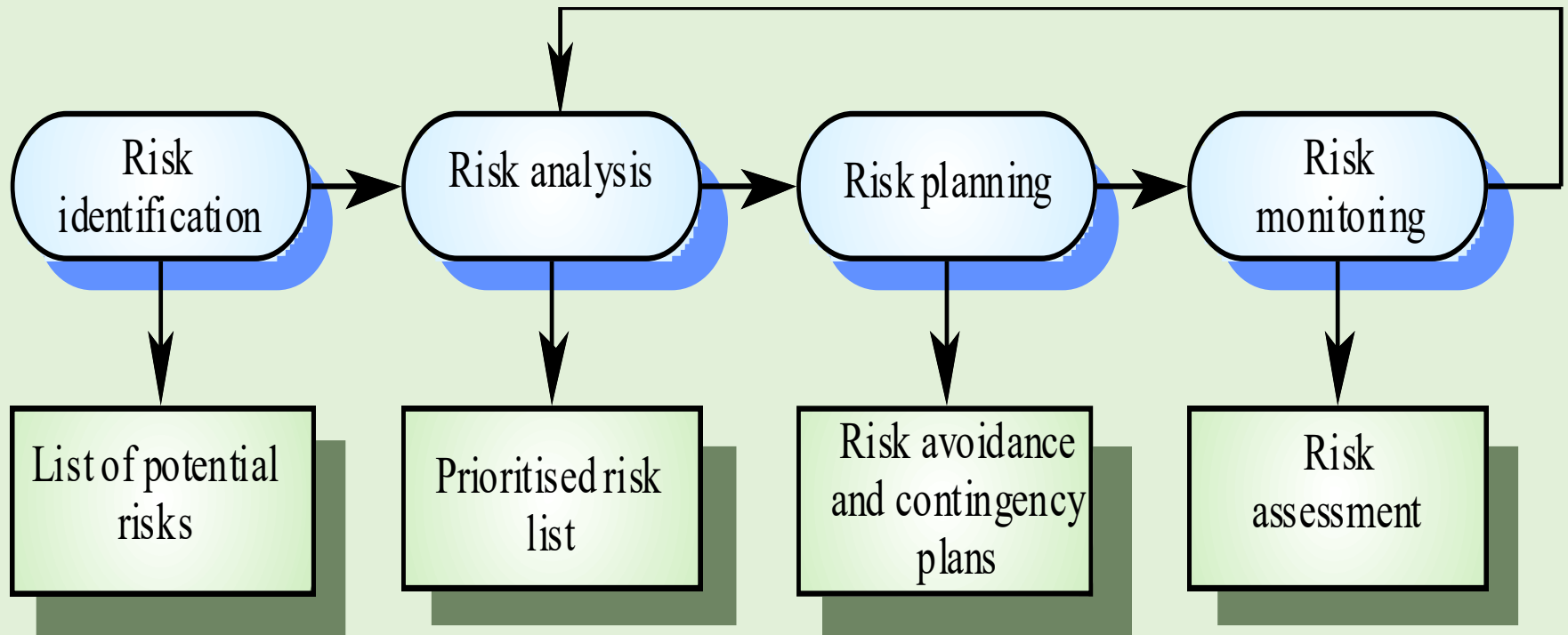
Risk Management

- **Risk management** is concerned with identifying risks and drawing up plans to minimise their effect on a project.
- A **risk** is a probability that some adverse circumstance will occur.
 - **Project risks** affect schedule or resources
 - **Product risks** affect the quality or performance of the software being developed
 - **Business risks** affect the organisation developing or procuring the software

Software Risks

Risk	Risk Type	Description
Staff Turnover	Project	Experienced staff leaves before the project is completed
Management Change	Project	Priorities change
Hardware Unavailability	Project	Not delivered on time
Requirements change	Project and Product	Lot of changes that will affect
Specification delay	Project and Product	Not available at the time of
CASE tools under performance	Product	CASE tools doesn't work as anticipated
Technology Change	Business	Technology changed
Product Competition	Business	Competitive product is marketed before the system is completed

The Risk Management Process



Risk Identification

- Technology risks
- People risks
- Organisational risks
- Requirements risks
- Estimation risks

Types of Risk

Risk type	Possible risks
Technology	The database used in the system cannot process as many transactions per second as expected. Software components which should be reused contain defects which limit their functionality.
People	It is impossible to recruit staff with the skills required. Key staff are ill and unavailable at critical times. Required training for staff is not available.
Organisational	The organisation is restructured so that different management are responsible for the project. Organisational financial problems force reductions in the project budget.
Tools	The code generated by CASE tools is inefficient. CASE tools cannot be integrated.
Requirements	Changes to requirements which require major design rework are proposed. Customers fail to understand the impact of requirements changes.
Estimation	The time required to develop the software is underestimated. The rate of defect repair is underestimated. The size of the software is underestimated.

Risk analysis

- Assess probability and seriousness of each risk
- Probability may be very low, low, moderate, high or very high
- Risk effects might be catastrophic, serious, tolerable or insignificant

Risk Analysis

Risk	Probability	Effects
Organisational financial problems force reductions in the project budget.	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project.	High	Catastrophic
Key staff are ill at critical times in the project.	Moderate	Serious
Software components which should be reused contain defects which limit their functionality.	Moderate	Serious
Changes to requirements which require major design rework are proposed.	Moderate	Serious
The organisation is restructured so that different management are responsible for the project.	High	Serious
The database used in the system cannot process as many transactions per second as expected.	Moderate	Serious
The time required to develop the software is underestimated.	High	Serious
CASE tools cannot be integrated.	High	Tolerable
Customers fail to understand the impact of requirements changes.	Moderate	Tolerable
Required training for staff is not available.	Moderate	Tolerable
The rate of defect repair is underestimated.	Moderate	Tolerable
The size of the software is underestimated.	High	Tolerable
The code generated by CASE tools is inefficient.	Moderate	Insignificant

Risk Planning

- Consider each risk and develop a strategy to manage that risk
- Avoidance strategies
 - The probability that the risk will arise is reduced
- Minimisation strategies
 - The impact of the risk on the project or product will be reduced
- Contingency plans
 - If the risk arises, contingency plans are plans to deal with that risk

Risk Monitoring

- Assess each identified risks regularly to decide whether or not it is becoming less or more probable
- Also assess whether the effects of the risk have changed
- Each key risk should be discussed at management progress meetings

Risk factors

Risk type	Potential indicators
Technology	Late delivery of hardware or support software, many reported technology problems
People	Poor staff morale, poor relationships amongst team member, job availability
Organisational Tools	organisational gossip, lack of action by senior management reluctance by team members to use tools, complaints about CASE tools, demands for higher-powered workstations
Requirements Estimation	many requirements change requests, customer complaints failure to meet agreed schedule, failure to clear reported defects

Key points

- A project **milestone** is a predictable state where some formal report of progress is presented to management.
- Risks may be **project risks, product risks or business risks**
- **Risk management** is concerned with identifying risks which may affect the project and planning to ensure that these risks do not develop into major threats

Examples

TASK	DURATION	DEPENDENCIES
T1	6	
T2	10	
T3	8	T1
T4	5	T2,T1
T5	8	T3
T6	15	T4,T5
T7	15	T1
T8	10	T6
T9	25	T6,T7
T10	10	T8,T9
T11	8	T10
T12	12	T11

Solution

- Give an example of a situation that entails uncertainty but not exposure, and hence no risk.

- Solution

- A spectator watches a man walk unprotected on a tightrope. The spectator is uncertain whether the man will fall, but is not personally exposed to a fall. Uncertain but not exposed, the spectator does not face risk.

Give an example of a situation that entails exposure but not uncertainty, and hence no risk.

- **Solution**

A farmer's crops die in the field due to a prolonged drought. The farmer is exposed to the drought, but he is not uncertain. The existence of the drought and its consequences are certainties to him. Exposed but not uncertain, the farmer does not face risk.

Cont...

- In our example of the deaths per passenger-mile risk metric, for what random variable's probability distribution may we interpret it as reflecting a mean?
- Solution
 - The random variable is the number of deaths that occur in a planned passenger-mile. The random variable can take on the values 0 or 1.